

Mounika Grandhi

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PROFESSIONAL SUMMARY

AI Engineer with 4 years of experience specialising in Generative AI, multi-agent systems, and RAG-based architectures. Proven record of architecting and delivering production-grade AI solutions using LangChain, LangGraph, DeepAgents, and MCP - reducing manual effort by 70% and improving AI response accuracy by 30%. Experienced in building end-to-end Agentic AI workflows, LLM orchestration across OpenAI, Anthropic, Gemini, and deploying scalable AI products on AWS and GCP.

SKILLS

Programming Languages: Python, Java

AI / ML Frameworks: LangChain, LangGraph, LangSmith, Google ADK, Deep Agents, LlamaIndex, FastMCP

Generative AI / LLMs: RAG, Agentic AI, Prompt Engineering, NLP, Machine Learning, Deep Learning, MCP, A2A

LLM Providers: OpenAI, Anthropic (Claude), Google Gemini, Grok

Databases & Vector Stores: MySQL, MongoDB, Milvus, ChromaDB, Vector Databases, Semantic Search

Cloud Platforms: AWS, Google Cloud Platform (GCP)

Backend & APIs: FastAPI, RabbitMQ, REST APIs

Tools & Version Control: Git, Docker, LLMops, CI/CD, Monitoring, HuggingFace

Soft Skills: Problem Solving, Cross-functional Collaboration, Agile Workflows

PROFESSIONAL EXPERIENCE

AI Engineer | Inncircles

Jul 2025 – Present

- Built a production-grade multi-agent AI framework using Deep Agents and LangChain, orchestrating multi-step specialized LLM agents with hierarchical sub-agent delegation and human-in-the-loop interrupts.
- Designed and implemented a RAG pipeline and Knowledge Base Search system using Milvus vector database, incorporating document chunking, semantic search, token-aware batching, grounding guardrails for output safety, and evaluation pipelines to measure retrieval quality and reduce hallucinations.
- Implemented multi-provider LLM orchestration across OpenAI, Anthropic, Gemini, and Grok with role-based model routing, structured output enforcement using Pydantic, token usage tracking, and intelligent fallback strategies for cost optimization - deployed as Dockerized FastAPI services on AWS with LangSmith for end-to-end monitoring and observability across providers.
- Integrated custom MCP tools by building a dedicated MCP server, applying AI to construction workflows and simplifying complex, multi-step operations into streamlined natural-language actions.
- Engineered a B2C WhatsApp AI assistant for construction project queries using hybrid RAG and LangGraph workflows, automating call scheduling and reducing manual effort for sales executives by 60%.
- Built a conversational analytics system for construction stakeholders using LangGraph agent workflows, reducing manual reporting effort by 50% via natural-language exploration of leads data with real-time insights.

Digital Specialist Engineer | Infosys

Aug 2022 – Jun 2025

- Engineered AI-driven code migration pipelines in the pre-ChatGPT era, translating legacy C/Java applications to Go using transformer-based models (Facebook's TransCoder) and OpenAI Codex (code-davinci-002) — with few-shot prompting, AST-based chunking, and unit-test validation.
- Pioneered an end-to-end code summarization pipeline leveraging LangChain summarization chains and language parsers, significantly improving content extraction and retrieval accuracy.

- Implemented RAG-based QA retrieval chains across 5+ legacy project codebases, enhancing efficiency and scalability in reverse engineering and knowledge extraction.
- Designed LLM evaluation pipelines using BLEU, CodeBLEU, and custom benchmarks to assess model output quality, accuracy, and hallucination rates ensuring robust post-migration validation.
- Crafted AI-powered Business Requirement Documents for legacy applications using LangChain, RAG, VectorDB, and embeddings, substantially enhancing document processing and knowledge retrieval.
- Refined prompt engineering strategies delivering a 30% improvement in AI-generated response accuracy and processing speed.

PROJECTS

Customer Support Assistant using LangGraph and Agentic AI

- Designed an AI-powered customer support assistant leveraging LangGraph and Agentic AI for dynamic, autonomous workflow execution, reducing manual support intervention through intelligent multi-agent routing.
- Deployed LLM-powered services on Google Cloud Platform using Vertex AI for model hosting and Cloud Functions for event-driven agent workflow execution.
- Integrated multi-agent retrieval mechanisms - vector database search, document retrieval for FAQs, database-powered product lookup via LangChain tools, and real-time web search enabling comprehensive, accurate responses across query types.

RAG Chatbot for Database with LlamaIndex

- Built a natural language interface for querying MySQL databases using LlamaIndex NLSQLTableQueryEngine and vector database semantic search, enabling non-technical users to extract insights without writing SQL.
- Implemented real-time SQL query generation via Groq LLM 70B model with HuggingFace embeddings for enhanced semantic query understanding and retrieval accuracy.

Swaram – Open-Source Voice AI Agent SDK

- Built an open-source TypeScript SDK for domain-agnostic voice support agents, orchestrating VAD, speech-to-text, LLM reasoning, tool calls, and text-to-speech without a hosted orchestration platform.
- Engineered real-time conversation flow with interruption handling, transcript tracking, and Twilio telephony transport, integrating Groq, Ollama, Whisper.cpp, and Kokoro TTS across the pipeline.

EDUCATION

Bachelor of Technology (B.Tech) — Information Technology	2018 – 2022
SRKR Engineering College, Bhimavaram, Andhra Pradesh	CGPA: 8.67
Class XII — MPC Narayana Junior College, Andhra Pradesh	2018 97.2%
Class X — SSC Narayana English Medium School, Andhra Pradesh	2016 CGPA: 10.0

CERTIFICATIONS

- AWS Certified Cloud Practitioner - Amazon Web Services
- Google Cloud Certified Associate Cloud Engineer - Google
- Salesforce Certified AI Associate - Salesforce
- Microsoft Certified: Azure AI Fundamentals - Microsoft
- Applied Generative AI Professional - Infosys
- Certified Python Programmer - Infosys
- Certified Mongo Developer - Infosys

AWARDS & RECOGNITION

- EnGenius Best Engineer Award - Infosys (Sep 2024 & Mar 2025)
- Rise Insta Award - Infosys (Mar 2024)